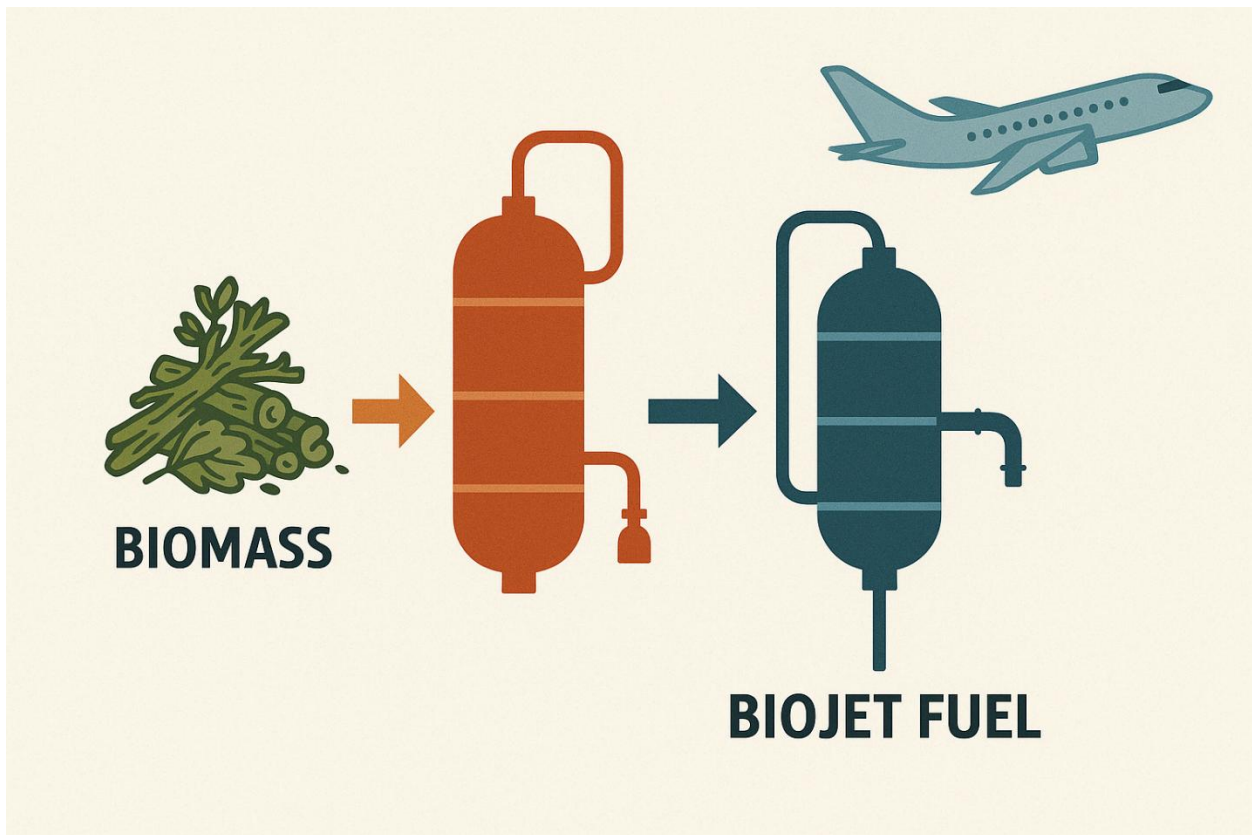


**Conversion of lignocellulosic biomass to jet fuel with a focus on
hydrodeoxygenation and reductive Catalytic fractionation.**



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ABSTRACT

The urgency towards decarbonizing the aviation industry has sparked significant interest in producing sustainable aviation fuels (SAFs) from renewable resources. Among the most promising strategies is the conversion of lignocellulosic biomass—particularly poplar wood—into jet fuel using the Reductive Catalytic Fractionation (RCF) followed by Hydrodeoxygenation (HDO). This paper explores the full scope of this pathway, from biomass structure and lignin chemistry to catalyst design and reactor configuration. By focusing on lignin-first valorization, the RCF process effectively extracts and stabilizes native lignin, which is then upgraded into jet-range hydrocarbons through selective deoxygenation. Our mass and energy balance analysis shows a progressive increase in fuel quality, with the final product reaching a heating value of 43.54 MJ/kg—comparable to conventional Jet A-1. A cradle-to-grave Life Cycle Assessment (LCA) further confirms the environmental benefits, with a global warming potential of just 0.28 kg CO₂e per kg of biomass processed. The results highlight not only the technical feasibility of this bio-jet production route but also its potential for real-world scalability. Key areas for improvement—such as catalyst stability, reactor efficiency, and feedstock engineering—are discussed to guide future research. Overall, this work offers a comprehensive, data-driven view of how lignin-first biorefining could contribute meaningfully to achieving net-zero aviation emissions.

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1.Introduction

Fossil fuels have become a vital source of energy, since the time they were discovered by human beings. At first early humans used fossil fuels like coal for cooking and heating purposes. It was not until the 1880's that humans used coal to generate electricity. Later on, people shifted towards petroleum which was then used to power colossal machines like planes, ships, etc. Thus as humans evolved, their usage of fossil fuels increased as well. Humans as a race have always traveled from one place to another in search of resources. Initially, humans were dependent on animal-powered vehicles, then came the era of steam engines and finally the internal combustion engine which used fossil fuels as a source of energy. Humans created multiple modes of transport via air, land, and sea routes. Out of these the fastest method to travel is via air. The first flight was invented by the Wright brothers and it took flight on December 17, 1903. This completely revolutionized the travel industry and led to the start of the aviation industry. The aviation industry is one of the fastest-growing industries in the world, expected to grow at an average rate of 4.3% by the ICAO (International Civil Aviation Organization) [1]. This would lead to an increase in the consumption of jet fuel. Most aviation fuel is conventionally supplied from kerosene produced from crude oil with an annual global consumption surpassing 330 Mt (2019) [2]. The aviation industry has helped humans advance rapidly but this growth comes at a cost. These fossil fuels produce massive amounts of greenhouse gases, particularly carbon dioxide (CO₂) which lead to global warming and several other adverse climatic effects. In 2019 the CO₂ emissions value reached 920 million tons, having increased approximately 30 percent since the past six years. The United States, with the world's largest commercial air traffic system, accounted for 200 million tons (23 percent) of the 2017 global CO₂ total [3]. The fuel costs represent about 24 % of the costs associated with the aviation sector[9]. This data was reported by the Environmental and Energy Study Institute (EESI). Many countries are now switching to cleaner alternatives, hence there is a push to produce sustainable aviation fuels (SAFs). The vision is to replace gasoline-fueled jets with electric or hybrid alternatives [3].

The main problem with producing jet fuels using sustainable sources is the composition of the fuel produced, unlike normal fuels used for vehicles, jet fuel has a unique composition. It has certain aromatic components without which it would not be able to support commercial airplane

engines. Jet fuel is produced by distilling crude oil ranging from 205 °C and 260 °C [5]. The jet fuel mainly consists of C₈ to C₁₆ hydrocarbons. The main components are alkanes, iso-alkanes, naphthene, and its derivatives and other aromatic compounds [6]. Each component adds a specific property to the fuel. The high hydrogen-carbon ratio of alkanes could ensure the energy density of the fuel. Naphthenes help to reduce the freezing point, which is of utmost importance in high-altitude flying. The aromatics contribute to the lubricating properties of the fuel to enhance the material compatibility and prevent leaks in the seals of an aircraft. Excessive aromatic components also influence the cleanliness of the fuel, so the content of aromatics should be controlled in a reasonable range [5,6,7,8].

Jet fuels can also be categorized based on their usage for example Jet A/Jet A-1 are kerosene based fuels which are used by commercial airlines. Jet B is a naphtha-kerosene based fuel used by commercial planes as well. On the other hand JP-4, 5, and 8 are used for military grade applications.

There are multiple alternatives to traditional jet fuel which are currently being investigated some of them are hydrogen-based fuels, electric aviation, solar-powered planes, liquified methane or propane gas. These alternatives

The data above helps us comprehend the need and urgency to synthesize biomass-based jet fuels or SAFs. The following paragraphs discuss the reasons as to why bio-jet fuels are required. Biomass can be divided into the following categories agricultural residues, forestry residues, algae, woody biomass, municipal and landfill waste. All of these categories consist of different materials in varying compositions. Biomass grows by using the carbon dioxide present in the atmosphere. This leads to the decarbonization of the atmosphere, and a cleaner future.

There are multiple methods that are being investigated for the production of biomass-based jet fuel. The properties and the process used depend on the kind of biomass we use. There are multiple pathways but generally the problem with biomass-derived jet fuel is that the feedstock itself is cheap but it requires multiple processing steps which increases the operating cost [10].

1.1. Hydroprocessing pathway: It involves the chemical conversion of triglyceride feedstock through hydrodeoxygenation, hydroisomerization, and hydrocracking to produce bio-based jet

fuel. Figure 1. Is a great representation of the entire process. First, the triglyceride feed is converted into a linear chain of hydrocarbons. This is done by hydrogenation of the triglyceride at high temperature and high pressure using a solid catalyst. The isomerization process converts the linear chain alkanes into C8 to C16 alkanes which are typically a large component of the composition of jet fuel. The distillation of this alkane mixture leads to bio-jet fuel and multiple other by-products. A 50% blend of commercial jet fuel and hydroprocessing pathway-based jet fuel has been utilized by many American airplane carriers [11].

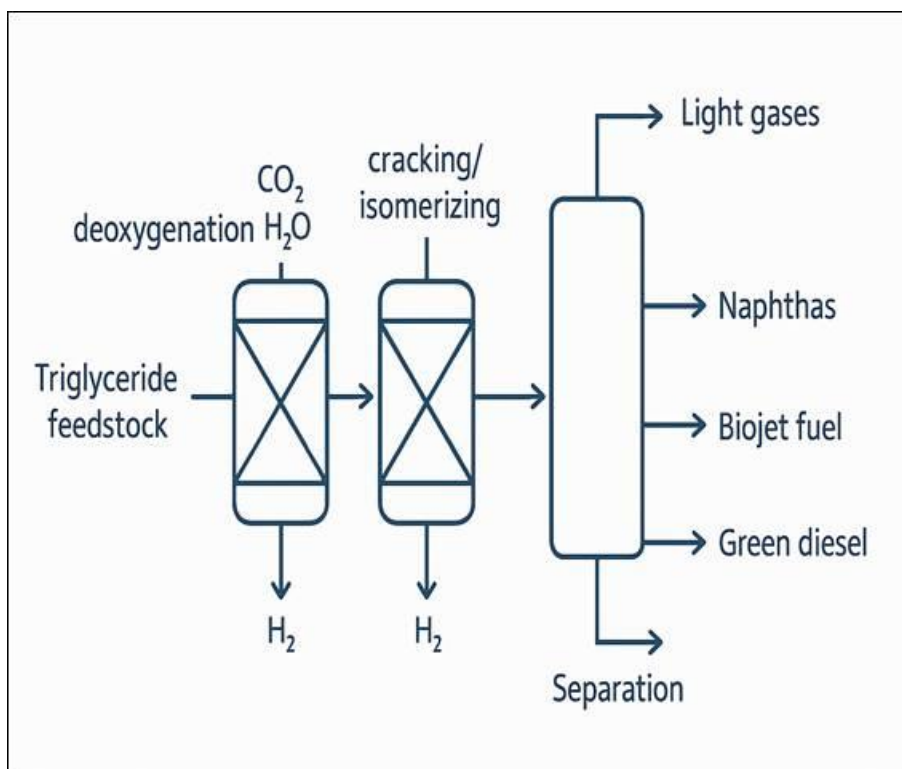


Figure 1: Process Flow Diagram for Hydroprocessing

1.2. Thermochemical Pathway: This pathway consists of processes which convert biomass into jet fuel using thermochemical processes like pyrolysis, gasification, and hydrothermal upgrading (HTU). Thermal gasification followed by the Fischer-Tropsch process is a well-known method to produce ASTM certified bio-jet fuel. The pyrolysis process is carried out in a single reactor, and the main objective is to maximize the liquid bio-oil production by using optimum temperature, residence and heating rate. [12]

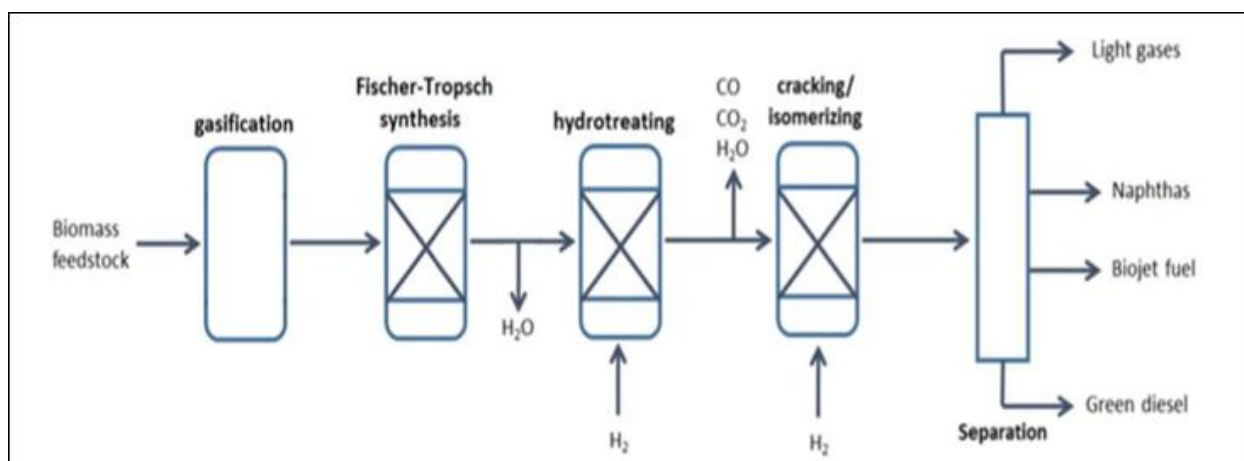


Figure 2: Process flow Diagram for Thermochemical Pathway

1.3. Oil from Algae: Oil from microalgae is a promising source of sustainable aviation fuel (SAF) due to its fast growth and high oil yield. Algae can be grown in diverse environments, including non-arable land and marine or freshwater systems, avoiding competition with food crops and reducing deforestation. The production process involves cultivating algae, extracting lipids from the biomass, and refining them into fuel through methods like hydroprocessing. This fuel can directly replace conventional jet fuel without requiring engine or infrastructure changes [14,15].

1.4. Alcohol to Jet Pathway: Sugars and starch-containing feedstocks can be converted into alcohol via fermentation. These alcohols can be converted into jet fuel by dehydrating them followed by oligomerization and hydrogenation. The fermentation can be done using enzymes or some kind of organisms making the process greener. The cost of fermentation of the sugars into alcohols is the largest cost in this process [12].

1.5 Hydrodeoxygenation (HDO) and Reductive Catalytic Fractionation (RCF): RCF is used to break the polymeric chains of lignin into monomers and these monomers are then converted into aromatics using HDO. These aromatics when in a proper composition could be used as jet-fuel. The RCF and HDO route is further discussed in detail in the sections below.

After discussing the most promising routes for the production of biomass-based jet fuels, the major drawbacks or challenges to the adoption of biomass-based jet fuels will be addressed.

There are multiple problems with the large-scale adoption of bio-jet fuels or SAFs. The economic feasibility of a process to produce SAFs needs to be assessed using Techno-Economic Analysis and the Life cycle assessment [13]. Optimization is to be done to reduce the number of steps required to treat the biomass feedstock. The biomass feedstock is heterogeneous in nature but for our reactions to occur we treat the feedstock and convert it into a homogenous mixture that can be used for the production of the jet fuel. Proper studies are to be done to understand the effect of product selectivity on factors like reaction time, catalyst, feedstock, and other reaction parameters [13].

The motivation behind this review is to drive the research in the field of biomass-based aviation fuels in a particular direction by compiling the existing information in the form of a review. The importance of biomass-based aviation jet fuel has already been discussed above. In this review, we will be focusing on the production of bio-aviation fuels from lignocellulosic biomass. The reasons behind choosing this feedstock are that it is available abundantly and is of low cost. We will be using the RCF (Reductive Catalytic fractionation) followed by Hydrodeoxygenation (HDO) route. This process was selected because of its novel approach, low carbon emissions, and employability. For a process to be employed on a large scale it should be easy to scale and simple enough to execute. Complexity makes it difficult to scale up a process. The fuel selected is JET A-1 . We are focusing on the production of this fuel because it is used heavily by commercial aviation carriers and would have a larger impact on the GHG emissions. Thus we hope to help fellow students and researchers understand the need for biomass-based aviation fuels.

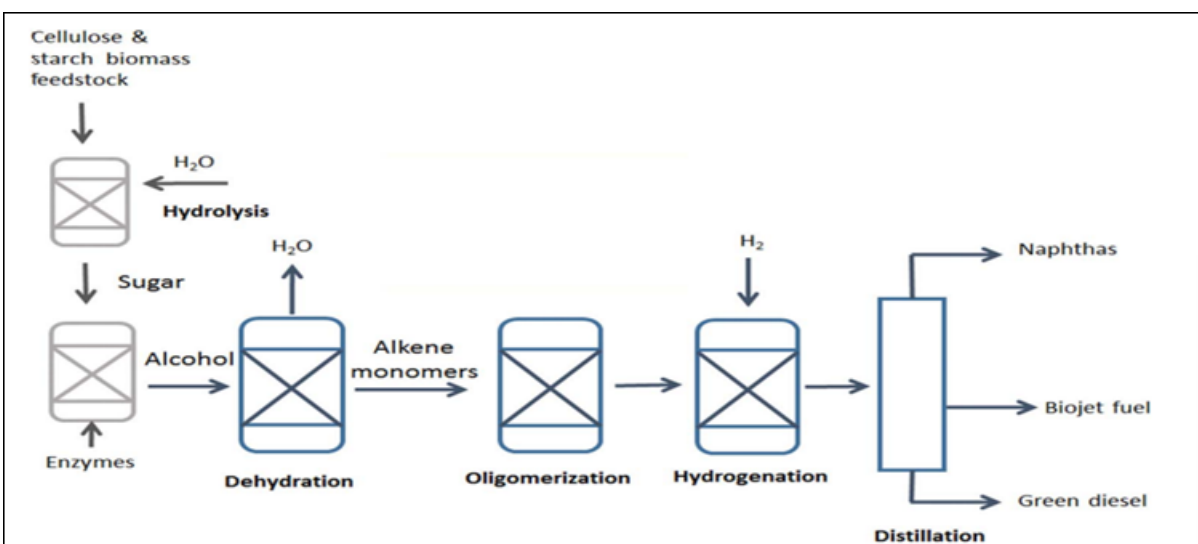


Figure 3: Process flow Diagram for Alcohol to jet Pathway

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2. LIGNOCELLULOSIC COMPOSITION

Lignocellulose, the largest portion of plants' structural material available on land, It consists primarily of three components: cellulose (40–60%), hemicellulose (10–40%) and lignin (15–30%). That combination can be found in woody plants like those of pine and birch, or the softer, herbaceous plants such as corn stalks and switchgrass. This structure is a long, linear polymer of how they are linked by β -1,4-glycosidic bonds, which give cellulose a strong, fiber-like structure. Hemicellulose, or, is a mixture of branching chains of carbohydrates composed of varying kinds of sugars that are generally chains varying in length from 50 to 300 units. Cellulose and hemicellulose, together with smaller sugars like pectins, comprise what is referred to as holocellulose, the carbohydrate fraction of plant biomass. Lignin is the most complex component of lignocellulose. It's an irregular and heavily oxygenated polymer that forms within the plant cell wall, when certain molecules, known as monolignols, undergo free-radical polymerization. This begins with the phenylpropanoid pathway, in which the amino acid phenylalanine is converted into three significant branches, namely, p-coumaryl alcohol, coniferyl alcohol, and sinapyl alcohol. The proportion of these compounds varies by type of plant—for instance, softwoods primarily consist of guaiacyl (G) units. The most abundant and well-known linkage type is the β -O-4 ether motif, which is most easily cleaved and usually the primary target of the depolymerization method. In contrast with the carbohydrate polymers, lignin also has a decent fraction of carbon-carbon inter bonds, such as 5-5, β -5, β -1, and β - β connections, whose relative amount is governed by monolignol distribution (S/G/H ratio). Thus, a relation can be elucidated by the fact that methoxy-substituted ortho positions cannot participate in forming 5-5 and β -5 carbon-carbon bonds. Thus, the S-units contain a lower percentage of carbon-carbon bonds compared to lignin, which consists of G-units. The extent of lignification is dependent on the origin of the plant. Softwoods have the highest percentage of lignin (21-25%), followed by hardwoods (18-25%) and herbaceous crops (15-24%).

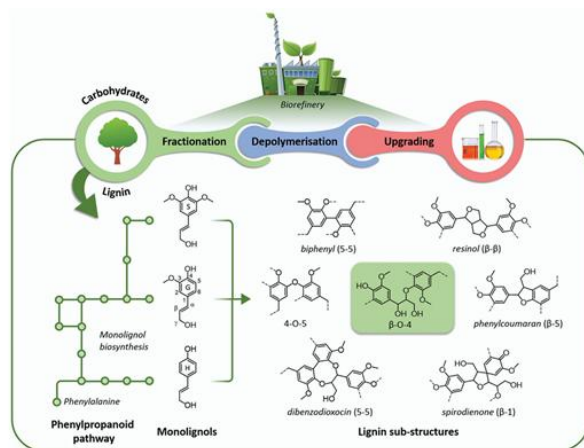


Fig. 1 The biosynthesis of lignin starts with the synthesis of monolignols through the phenylpropanoid pathway, followed by translocation to the cell wall where radical polymerisation takes place, leading to various ether and carbon-carbon inter-unit bonds. The successful utilisation of this complex bio-polymer as a feedstock for chemicals is governed by an interplay of (i) the biomass fractionation method, (ii) the lignin depolymerisation technology, and (iii) subsequent upgrading towards targeted chemicals. These three interconnected stages form the backbone of the following review.

Figure 4: Lignin Biosynthesis [16]

3. Recalcitrant resource to Jet Range Products

Fractionation is often the primary step in the refinery to reduce the complexity of the lignocellulosic biomass. Adequate depolymerisation followed by chemical or biocatalytic upgradation is necessary for formulating a lignin valorisation chain. Fractionation, Depolymerisation, and upgrade are the three traditional sequential steps in the valorisation chain. There are a few processes that are performed in a single operation. Reductive catalytic fractionation is an emerging technology that combines fractionation and reductive depolymerisation stabilization, leading to the formation of depolymerised lignin oil. The other advantage of this method is that it avoids the degradation of lignin, which is a common problem in conventional lignin processing techniques

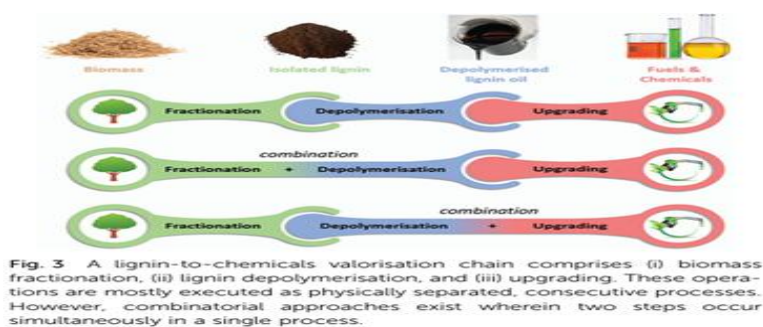


Fig. 3 A lignin-to-chemicals valorisation chain comprises (i) biomass fractionation, (ii) lignin depolymerisation, and (iii) upgrading. These operations are mostly executed as physically separated, consecutive processes. However, combinatorial approaches exist wherein two steps occur simultaneously in a single process.

Figure 5: Lignin Valorisation1 [16]

4. Lignin Extraction

Conventional bio-refineries use acidic or basic extraction conditions at elevated temperatures. Under these conditions, lignin undergoes deleterious reactions where cleaved β -O-4 linkages form unstable intermediates that form C-C bonds. In the case of Kraft, lignosulfonate, and soda lime processes produce degraded technical lignin with β -O-4 linkages less than 20%, whereas native lignin contains β -O-4 contents between 45% and 85%. Such bond profiles require aggressive upgradation methodologies, resulting in the formation of products that are too volatile for aviation fuel and demanding further upgradation, for example, alkylation. Alternative methods reported that yield like native lignin with β -O-4 linkages is obtained from mild lignin extraction conditions with organic and dilute acid solvents, but unfortunately, the extent of lignin extraction attained from this scenario is low. To get a higher extent of extraction with the native lignin grade, flow-through extraction with short residence time was proposed as an alternative to batch methods.

Lignin first strategies address this challenge by stabilising reactive intermediates, allowing efficient lignin extraction without any C-C coupling reactions. The other standard methodology for extracting native lignin is biological processing that involves enzymatic treatment targeting specific sugars under mild conditions, which yields lignin-rich biomass. Regardless of the extraction methodology, the consensus is that successful lignin upgradation hinges on targeting the C-O during depolymerisation, due to its low bond dissociation energies (49-72 Kcal/mol) relative to C-C bonds (101-118 Kcal/mol). This allows depolymerisation to retain much of the inherent structure while yielding low molecular weight compounds after depolymerisation.

5.0 Oxidative depolymerization

A common depolymerization targets the aryl ether bonds of the lignin structure. While showing nearly the theoretical equivalent in lignin yield, the products resulting from this methodology give products like phenyl alkanoic acids(C7-C8), which, after hydrotreatment, do not align with the range of jet fuels from C8-C16. Thus, this type of depolymerisation is aligned with commodity

chemical generation or as a precursor for bioconversion rather than a precursor for fuel upgradation.

6.0 Reductive depolymerisation

Reductive catalytic fractionation (RCF) is a promising method for extracting and depolymerizing native lignin and preventing lignin condensation. This method reduces the lignin by the methods of solvolysis and hydrogenolysis while making the reactive intermediates stable by utilization of reductants and redox catalysts. The pathway of solvolysis or hydrogenolysis depends on different parameters like pH, solvent or catalyst. RCF is not typically effective at cleaving the C-C bonds and is thus reliant on aryl ether bonds. This pathway forms H, S, G monomers with C1-C3 side chains and end group content. Other than extraction and depolymerisation RCF uses aldehydes and diols to protect the reactive intermediates formed during fractionation. Beyond combined extraction and depolymerisation obtained via RCF, both aldehydes and diols have been used to protect reactive intermediates formed during fractionation.

6.1. Experimental versus theoretical combustion energy retention

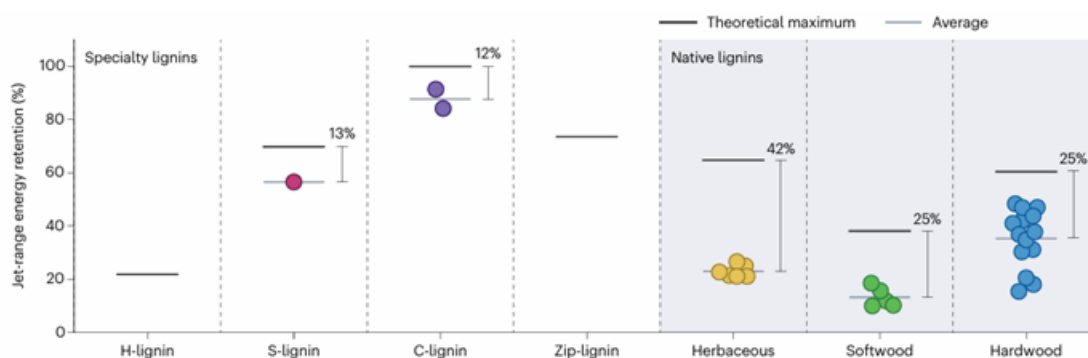


Fig. 3 | Lignin extraction/depolymerization strategies. Combustion energy retentions reported for the reductive depolymerization of a variety of different lignins compared with theoretically achievable values based on reported β -O-4 contents and the assumption that the theoretical monomer yield is equivalent to the square of the lignin ether bond content^{33,36,161}. Each point represents an enthalpic yield for a given study; solid black lines represent the theoretical limit

based on ether content (Table 1); and solid grey lines represent the average of the reported literature values. LHV's were calculated using the reported monomeric compositions along with the modified Dulong formula¹⁵¹. Lignin monomer yields were extracted from the relevant literature^{77,90,91,93,94,98-100,162-164}. Calculation details can be found in Supplementary Note 3 and Supplementary Table 7.

Figure 6 : Lignin Extraction [17]

The above figure shows the theoretical and documented monomer energy retention values during the depolymerisation of various lignin's. In all the scenarios experimental enthalpic yields are lower than theoretical enthalpic yields, the extent of disparity depends on the feedstock.

Various studies have investigated the influence of feedstock on lignin monomer yield under same RCF conditions. Typically, monomer yields follow the sequence hardwood > herbaceous biomass > softwood. Lower β -O-4 contents in softwood lignins relative to hardwoods are the reason behind decreased monomer yield. Herbaceous lignins, though showing ether contents similar to hardwoods, produce lower yields. The reason behind this is the amount of lignin-carbohydrate linkages in grasses which may hinder delignification.

Alternative natural and genetically modified lignins show promise for high combustion energy retention. S-lignin, rich in S-units and β -O-4 linkages, has a theoretical enthalpic yield of 73%, with experimental retention around 57%. C-lignin, found in limited plant sources, offers nearly 100% theoretical energy yield due to its benzodioxane structure and experimentally shows 84–91% retention. Genetically engineering plants to produce more C-lignin could boost aviation fuel production. Zip-lignin also shows potential due to its ester linkages, though more research is needed on its catalytic depolymerization.

Monomer and energy yields in biomass processing also depend on reactor design. While batch reactors are common for high monomer yields, they face issues like catalyst and pulp recovery. Flow-through reactors offer advantages but require high solvent use, which may be reduced through reactor innovations like effluent recycling. Another approach is using in situ hydrogen donors (e.g., solvents or sugars) instead of external hydrogen, which can lower costs and environmental impact. However, this may lead to lower carbon yields due to CO₂ release. Some studies have explored RCF without using external hydrogen.

7.0 Upgradation of Lignin-derived Bio-oils

7.1. Hydrodeoxygenation

Alternative natural and genetically modified lignins offer promising avenues for achieving higher energy retention during biomass upgrading. Among them, S-lignin—characterized by its abundance of syringyl (S) units and β -O-4 linkages—shows a theoretical enthalpic yield of 73%, while experimental studies report around 57% energy retention. In contrast, C-lignin, which occurs in limited plant species, displays an exceptionally favorable benzo-dioxane structure that

supports nearly 100% theoretical energy yield. Experimental data shows C-lignin achieving energy retention between 84–91%, making it a strong candidate for biofuel production. Increasing the expression of C-lignin in plants through genetic engineering could greatly increase the production of sustainable aviation fuel. Although more research is required, Zip-lignin, another designed variation, has ester connections that are simpler to break down, providing more potential in catalytic depolymerisation.

Reactor design also plays a crucial role in determining monomer and energy yields during biomass processing. Batch reactors are commonly used due to their ability to provide high monomer yields, but they present challenges such as complex catalyst recovery and pulp separation. Flow-through reactors, by contrast, allow for better separation and catalyst reuse; however, they require larger volumes of solvents. Improvements like effluent recycling systems help mitigate this issue and improve the efficiency of the process. Additionally recent strategies favour using insitu hydrogen donors such as alcohols in place of external hydrogen gas. This shift reduces both cost and environmental impact, although it may decrease carbon yields due to the release of CO₂. Some studies have demonstrated successful Reductive Catalytic Fractionation (RCF) without external hydrogen, further advancing the sustainability of biomass valorization.

The relevance of these lignin types and process strategies is well illustrated in Figure 4. The illustration below illustrates how the aromatic monomer structure affects the thermodynamic profiles of various deoxygenation processes, particularly direct deoxygenation (DDO) and indirect deoxygenation (IDO). As shown in Figure 4b–d, the favorability of hydrogenation stages in the IDO pathway is changed when functional groups such as hydroxyl or methoxy are present on the aromatic rings. Significantly, the chemical homogeneity of C-lignin results in more consistent deoxygenation behaviour, frequently favouring DDO, which preserves the aromatic ring and optimises energy retention. These thermodynamic discoveries highlight how crucial it is to select the ideal lignin feedstock and catalytic pathway in order to improve energy efficiency and direct the development of new biofuel production methods.

Figure 7: Catalyst Design Consideration [17]

7.2. Catalyst Design Beyond Thermodynamics

Thermodynamics is needed for the estimation of conversion of feedstocks in the HDO process but they are not the only consideration while adjusting product selectivity. In order to overcome kinetic barriers that hamper the reaction rate, industrial hydrogenation occurs at the temperature greater than 300C. Thus, enhancing the selectivity and efficiency of these processes is majorly dependent on design and engineering the catalyst. The goal of research has been to maximise carbon efficiency while minimising side products using the catalytic deoxygenation of lignin-derived model compounds such as phenol, guaiacol, and anisole.

7.3. Effect of Metal Type on Activity and Selectivity

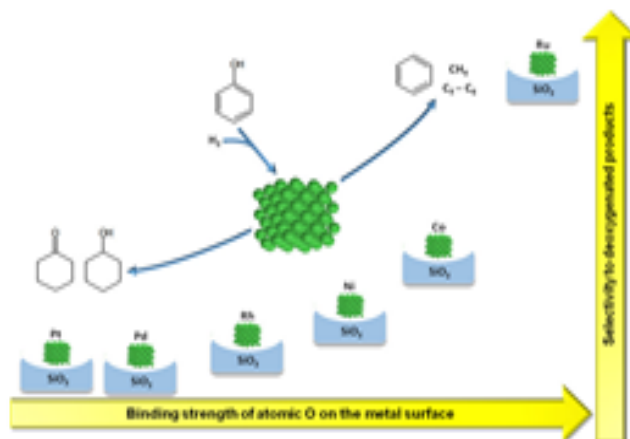


Figure 8: Effect of Metal Type [17]

The choice of metal significantly influences HDO activity and the reaction pathway that is direct de-oxygenation or hydrogenolysis. Supported metal catalysts, especially on inert silica or carbon supports have been extensively studied to isolate the effect of metal itself. One important study demonstrated the deoxygenation activity of phenol over inactive SiO₂ support, following the trend: Ru>Co>Ni>Rh>Pd>Pt. This trend correlates with the oxophilicity of the metal, where more

oxophilic metals like Ru, Co, and Ni favor C-O cleavage and hence promote deoxygenation. In contrast, less oxophilic metals like Pt and Pd favor aromatic ring hydrogenation, limiting their role in selective deoxygenation processes. The stronger the bond between the metal and the oxygen-containing functional groups facilitate the elongation of C-O bond, making the cleavage easier and more efficient.

7.4. Role of Catalyst Supports and Oxophilicity

Beyond the metal type, the catalyst support plays an equally important role in influencing catalyst performance, particularly in terms of metal dispersion, adsorption of intermediates, and product desorption. Supports with high oxophilicity, such as certain metal oxides, have been shown to improve the selectivity of deoxygenated products significantly. Studies on Pd incorporated supports report a selectivity trend for HDO of phenol and anisole: $\text{Nb}_2\text{O}_5 > \text{TiO}_2 > \text{ZrO}_2 > \text{SiO}_2 > \text{CeO}_2$. This order shows the capability of the support to make the oxygen-containing intermediates stable, improve adsorption strength, and promote bond cleavage. Techniques like diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS) have confirmed that highly oxophilic supports account for improving adsorption and desorption dynamics, especially in the removal of hydroxyl groups, which is typically even tougher to cleave in comparison with methoxy groups due to their bond dissociation energy.

7.5. Metal Support Interactions and Functional Roles

The interaction between the metal and its support can significantly alter reaction pathways, especially for the direct deoxygenation (DDO) mechanism. For DDO to proceed efficiently, the catalyst requires a combination of features: an active metal site, a nearby oxygen vacancy, and a reducible support that can aid in substrate activation. The criteria are better met with acidic and oxophilic supports than with inert ones. For example, Pd supported on Nb_2O_5 showed a marked increase in deoxygenation selectivity during the HDO of phenol compared to Pd/ SiO_2 . Similarly, the performance of Pt on H form beta zeolite (H-BEA) versus Pt on SiO_2 for anisole conversion showed that while demethylation occurred on metal center in both cases, the support still subtly influenced the extent of reaction. From these observations, it is evident that metal support

interactions are not always independent of the functional group involved and may vary depending on the substrate.

7.6. Bimetallic Catalysts and Metal-Metal Interactions

In addition to metal support interactions, bimetallic catalysts offer another avenue to enhance HDO performance. These catalysts benefit from synergistic effects due to electronic and geometric modifications that go beyond the capabilities of single metal systems. The inclusion of a second metal changes surface bond lengths, bond angles, and even electronic properties of active sites, responsible for improved reactivity and selectivity. Moreover, metal dopants in bimetallic systems often enhance catalyst stability by reducing carbon deposition and preventing the sintering of active metal centers. However, designing optimal bimetallic combinations and understanding their active sites' nature remain complex due to the structural intricacies involved.

7.7. Reactivity discrepancies between models and real feedstocks

There is an extensive number of studies using model compounds for lignin to gain insights into HDO. It is equally important to understand the catalyst and process performance associated with real lignin-derived bio-oils. These effects can be partially seen in model studies where primary, binary, and ternary mixtures of compounds are used for HDO. **Teles** used a series of supported Pd catalysts to deoxygenate binary phenolic mixtures to reveal that hydroxyl groups selectively adsorb on catalysts relative to methyl ethers. This is responsible for far higher deoxygenation rates in phenols and cresol relative to anisole, affecting the product selectivity at kinetically limited conditions. This is an evidence that examining competitive adsorption in studies using complex feedstocks, as disparity in reactivity can result from differences in functional group steric and electronic effects. Aside from adsorption effects, increased catalyst deactivation can also result in reactive differences between model compounds and lignin substrates.

7.8. Active Site Blockage

It is one of the primary causes of catalyst deactivation during the upgradation of Biomass. It happens when components of the reaction mixture (for instance, carbon (coke), salts, heteroatoms, and so on) deposit on the catalyst surface, thereby limiting substrate accessibility. For example, Cordera-Lanzac et al reported the primary pathways attributed to the deactivation of Pt-Pd bimetallic catalyst for the HDO of black poplar-derived fast pyrolysis oil. One pathway involved the recombination of oxygenated components and the condensation of post-HDO alkyl aromatic hydrocarbons, common coke precursors. In general, the presence of metal centres diminishes coke formation, which can be attributed to their ability to oxidise coke at low temperatures and promote hydrogen spillover, which reverses the influence of hydrogen abstraction on coke formation.

7.9. Active site Loss

It is another common mode of catalyst deactivation. The mechanism of this kind of deactivation is dependent upon the nature of the active site (metallic, acidic), thermodynamic driving force (applied potential, temperature). One example of active site loss is leaching, where the active site is dissolved in the reactant solution. For example, Shafaghat et al noticed the aqueous HDO efficiency with repeated usage, accompanied by decreased catalyst metal loading. This suggests that catalyst leaching is the primary cause.

7.10. Active site modification

This can occur due to oxidation or reduction of the catalyst's active components or their interaction with feed impurities. This changes the chemical state of the catalyst and thus its reactivity. For example, Kumar et al. showed that oxidizing the surface of Mo_2C altered its selectivity during anisole HDO, shifting products from cyclohexane toward phenol. Conversely, Prasomsri et al. found that reducing molybdenum oxide from Mo^{6+} to Mo^{4+} led to a significant decrease in its HDO activity for aromatic model compounds derived from lignin.

Overall, although model systems are useful for understanding fundamental thermodynamic and kinetic aspects of HDO, the complexities introduced by real feedstocks—including diverse functional groups, monomer and oligomer mixtures, and various heteroatomic impurities—

necessitate their inclusion in catalyst testing. Issues like active site competition and catalyst deactivation are far more pronounced with real bio-oils, and overlooking them may limit the practical applicability of findings derived solely from model compound studies.

8.0. Mass Balance and Energy Balance Calculations

We have discussed the processes that could be employed to produce bio-jet fuel in the section above. This section would focus on a route which holds the most potential when it comes to making SAF's and an economically viable product. The route has been discussed in detail in the section above. The route follows a method where the raw biomass, which consists of lignin, is converted into monomers via Reductive Catalytic Fractionation (RCF) using Nickel supported on carbon as a catalyst. The temperature range at which this is carried out is around 350 to 400 °C.

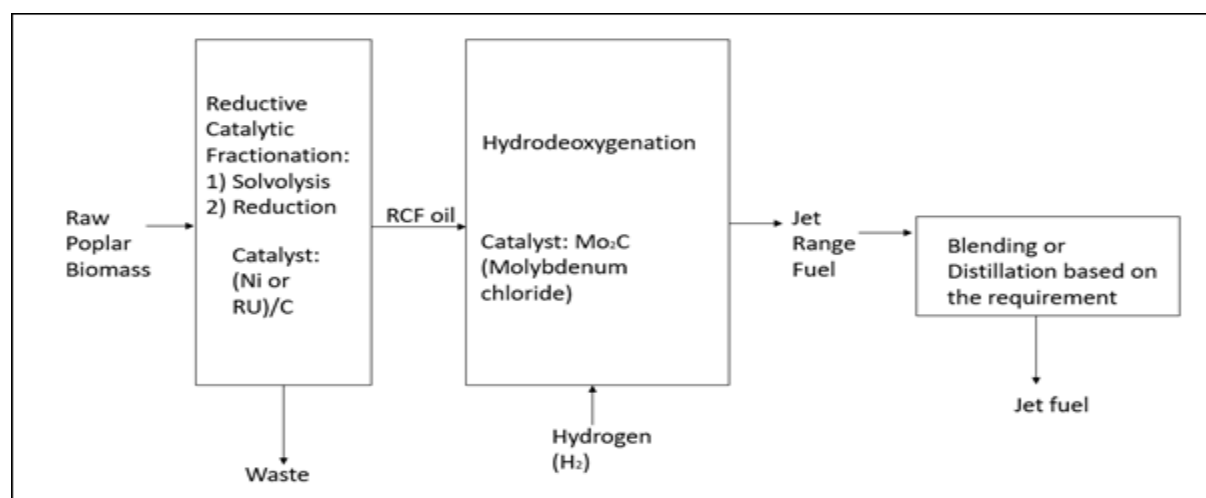


Figure 9: Process flow Diagram

These monomers are called as RCF oil. This RCF oil is then subjected to hydrodeoxygenation (HDO), where Mo_2C is used as the catalyst to convert it into a jet fuel grade oil [18]. This oil could be further distilled in order to reach a specific composition. The reasons for using poplar biomass are that poplar trees are ready to harvest within two to three years after planting. The trees do not need replanting because the new plant regrow from the shoots left after the harvesting. Thus, this ensures multiple harvesting cycles and reduces the cost of the raw cellulose feedstock [2]. Poplar is a hardwood tree, and hybrid poplar trees could also be created using controlled

pollination. Poplar requires less labour, irrigation, fertilisers and has higher resistance to pests than other traditional energy crops like corn, etc [19].

In this section, we will focus on the energy and mass balance calculations associated with the production of bio-jet fuel using the aforementioned process.

Stream no.	Stream description
1	Input to the RCF reactor
2	Output of the RCF reactor/Input to the HDO reactor
3	Output of the HDO reactor
4	Waste Stream of the RCF reactor
5	Input of the Blending/distillation column
6	Output of the Blending/distillation column

Table 1: Stream Descriptions of the PFD

The biomass obtained from any source generally consists of about 15 to 30% lignin, the rest consists of cellulose, hemicellulose, etc. Lignin is a compound that can be further converted into useful products. Lignin is generally used in most modern biorefineries as a source of low-energy fuel, but it can be valorized using methods like thermochemical or biological funelling processes [20,21,22].

The RCF process involves two steps, solvolysis and reduction. This helps us reduce the kinds of monomers formed from the β -O-4 bond cleavage. Typical temperatures used range from 180 °C to 250 °C for 2–12 hr, at reaction pressures of 5–15 MPa, using a solid catalyst mixed with biomass in a 0.5–2:10 mass ratio, a polar protic solvent such as alcohol or water, and a reductant (e.g., H_2). Any temperatures greater than this result in the solubilization of the carbohydrates, which leads to the formation of multiple products affecting the composition of the feed [23,24,24].

Compound	Average (wt frac.)
Methylcyclohexane	0
Toluene	0.0017

Propylbenzene	0
Alkylated Propylbenzene	0.0007
Propylcyclohexane	0.001
Propylphenol	0.0017
Alkylated Propylphenol	0.0017
Methoxy Propylbenzene	0
Propylguaiacol	0.13
Rearranged Propylguaiacol	0.002
Alkylated Propylguaiacol	0.0042
Propylsyringol	0.243
Propanol Guaiacol	0.051
Propenyl Guaiacol	0.002
Propanol Syringol	0.086
Propenyl Syringol	0.002
Monomer Total	0.512
Oil Total	0.469
Mon/Oil	0.5215
Total	0.981 +/- 0.02

Table 2: RCF Output feed composition [25]

The table above shows us the composition of the lignin oil in weight fraction. The calculations below will be done on the basis of this. Hydrodeoxygenation was carried out at 325 to 400 °C using a Mo₂C catalyst. The table below consists of the different arrangements of temperatures and passes used for the HDO study in paper [9]. The table compares the composition of the products obtained via HDO.

Category	400°C Single-Pass	375-375°C Dual-Pass	350-375°C Dual-Pass	325-375°C Dual-Pass
Aromatic	77.2	82.8	80.4	80.1

n-Alkane	1.3	1.4	1.3	1.5
Iso-alkane	0.3	0.6	0.4	0.1
Cycloalkane	5.9	9.8	7.3	7.2
Alkene	0.7	0	0	0
Oxygenate	3.3	0.1	3.4	3
Unidentified	0	0	0	0
Total Mass % Accounted	88.7	94.7	92.8	91.8
Aromatic Breakdown	77.2	82.8	80.4	80.1
Benzene	3.3	3.6	3.3	1.2
Toluene	3	4.7	4	3.1
Xylene	3.1	3.2	2.9	2
n-Propylbenzene	43.3	45.8	44.6	46
Alkylbenzene	12.8	12.6	11.8	10.9
Indene	1.9	1.3	0.2	0.6
Tetralin	0	0	0	0
Naphthalene	0.7	0	0.1	0.1
Biphenyl	8.5	9.5	11.9	14.2
2-ring	11.8	12.8	13.6	16.5

Table 3: Composition of the HDO product stream [25]

The information provided in table 1 and table 2 would be used to perform the mass balance for a biomass feed of 100 kgs. Assumption: out of the raw biomass feed stream 22.5% is lignin. For calculations, the 375-375 °C data has been used because it has the highest value of conversion.

For the RCF unit	
Mass in	Mass out

Components	stream 1	units	Components	stream 2	units
Raw biomass	100	kg	Methylcyclohexane	0	kg
Lignin	22.5	kg	Toluene	0.03825	kg
			Propylbenzene	0	kg
			Alkylated Propylbenzene	0.01575	kg
			Propylcyclohexane	0.0225	kg
			Propylphenol	0.03825	kg
			Alkylated Propylphenol	0.03825	kg
			Methoxy Propylbenzene	0	kg
			Propylguaiacol	2.925	kg
			Rearranged Propylguaiacol	0.045	kg
			Alkylated Propylguaiacol	0.0945	kg
			Propylsyringol	5.4675	kg
			Propanol Guaiacol	1.1475	kg
			Propenyl Guaiacol	0.045	kg
			Propanol Syringol	1.935	kg
			Propenyl Syringol	0.045	kg
			Oil Total	10.5525	kg
			Total Mass	22.41	kg
			Waste	77.59	kg

Table 4: Mass balance for the RCF unit

For the HDO unit					
Mass in			Mass out		
Components	stream	unit	Components	stream	unit
	2	s		3	s

Methylcyclohexane	0	kg	Aromatic	18.5554	
Toluene	0.03825	kg	n-Alkane	8	kg
Propylbenzene	0	kg	Iso-alkane	0.31374	kg
Alkylated Propylbenzene	0	kg	Cycloalkane	0.13446	kg
Propylcyclohexane	0.01575	kg	Alkene	2.19618	kg
Propylphenol	0.0225	kg	Oxygenate	0	kg
Alkylated Propylphenol	0.03825	kg	Unidentified	0.02241	kg
				0	kg
Methoxy Propylbenzene	0	kg	Total Mass % Accounted	21.2222	
				7	kg
Propylguaiacol	2.925	kg	Aromatic Breakdown	18.5554	
Rearranged Propylguaiacol	0.045	kg	Benzene	8	kg
Alkylated Propylguaiacol	0.0945	kg	Toluene	0.80676	kg
Propylsyringol	0.045	kg	Xylene	1.05327	kg
	5.4675	kg		0.71712	kg
Propanol Guaiacol				10.2637	
	1.1475	kg	n-Propylbenzene	8	kg
Propenyl Guaiacol	0.045	kg	Alkylbenzene	2.82366	kg
Propanol Syringol	0.045	kg	Indene	0.29133	kg
Propenyl Syringol	1.935	kg	Tetralin	0	kg
Oil Total	0.045	kg	Naphthalene	0	kg
Total Mass	10.5525	kg	Biphenyl	2.12895	kg
Waste	22.41	kg	2-ring	2.86848	kg
	77.59	kg			

Table 5: Mass balance for the HDO unit

For the energy balance calculations, the lower heating values and higher heating values have been calculated for the input feed stream and the outlets of the RCF and HDO units.

For the input feed we use the equation: $HHV = 0.089 \cdot L + 16.82$ [26] (where L is the wt% of lignin).

The HHV obtained is **18.8225 MJ/kg** for the feed biomass.

Equation No.	Formula	Reference
1	$HHV = 0.326C + 3.46$	[27]
2	$HHV = 0.335C + 1.423H - 0.154O - 0.145N$	[28]
3	$HHV = 0.349C + 1.178H + 0.100S - 0.103O - 0.015N$	[29]
4	$HHV = 0.00355C^2 - 0.232C - 2.23H + 0.0512C \times H + 0.131N + 20.6$	[30]

Table 6: Equations used to calculate the HHV values for the RCF and HDO unit product streams

RCF oil							
Equation	C	H	N	O	S	HHV (MJ/kg)	References
1	65.44	7.93	0.19	26.53	0	24.79	[27]
2	65.44	7.93	0.19	26.53	0	29.09	[28]
3	65.44	7.93	0.19	26.53	0	29.44	[29]
4	65.44	7.93	0.19	26.53	0	29.53	[30]
Average						28.21	

Table 7: RCF oil HHV values calculated using different equations

HDO Products							
Equation	C	H	N	O	S	HHV (MJ/kg)	References
1	87.69	11.28	0.005	1.025	0	32.04	[47]
2	87.69	11.28	0.005	1.025	0	45.26	[48]
3	87.69	11.28	0.005	1.025	0	43.78	[49]
4	87.69	11.28	0.005	1.025	0	53.04	[50]
Average						43.54	

Table 8: Calculated HHV values for the HDO unit product stream

From these calculations, you can observe that the heating value for the product streams increases after each process. The heating value of the RCF oil is 54.83% greater than that of the raw biomass feed. The final product stream (jet fuel) has a heating value that is 53.28% greater than the RCF oil. Thus to reach the conclusion that this process works the calculated heating value of the jet fuel obtained via the above-mentioned route was compared with commercial Jet-A fuel.

The heating value of commercial jet-A fuel is 43.40 MJ/kg, whereas for our bio-jet fuel, the value is 43.54 MJ/kg. The values are extremely close and thus make RCF followed by HDO a very attractive route to produce bio-jet fuel. Addition of certain sulphur compounds needs to be done, or blending is required to reach the perfect composition as that of commercial jet fuel. This can be easily achieved using a blending unit. The next section discusses the economic and LCA analysis of biomass based jet fuel.

9. LIFE CYCLE ASSESMENT OF THE BIOFUEL DERIVED FORM THE RCF METHOD

Reductive Catalytic Fractionation (RCF) is a lignin-first biorefinery strategy that integrates biomass fractionation and lignin depolymerization with simultaneous catalytic stabilization. This process valorizes both lignin and carbohydrate fractions from biomass such as poplar, allowing the production of high-value phenolics and fermentable sugars for biofuels. RCF distinguishes itself from traditional methods by minimizing lignin condensation and maximizing monomer yields through reductive stabilization in alcohol solvents and optional H₂-free operation [51].

The LCA model in this study was constructed in OpenLCA 2.4 using free databases from openLCA Nexus including LCA Commons and USLCI, alongside customized process data based on the Stone et al. Joule (2022) RCF-HDO model [53] and Budsberg et al. (2016) fermentation and ATJ pathway [54]. The model evaluates the full process from poplar cultivation to SAF combustion.

9.1. Goal & Scope of the LCA

- **Goal:** Evaluate cradle-to-grave environmental impacts for 1 kg poplar biomass processed via RCF, fermented to ethanol, converted to sustainable aviation fuel (SAF), and combusted in flight.
- **Scope:** Includes poplar cultivation, biomass transport, RCF, fermentation, Alcohol-to-Jet (ATJ) upgrade, and jet fuel use.
- **Functional Unit:** 1 kg dry poplar biomass
- **Impact Method:** ReCiPe Midpoint (H) 2016

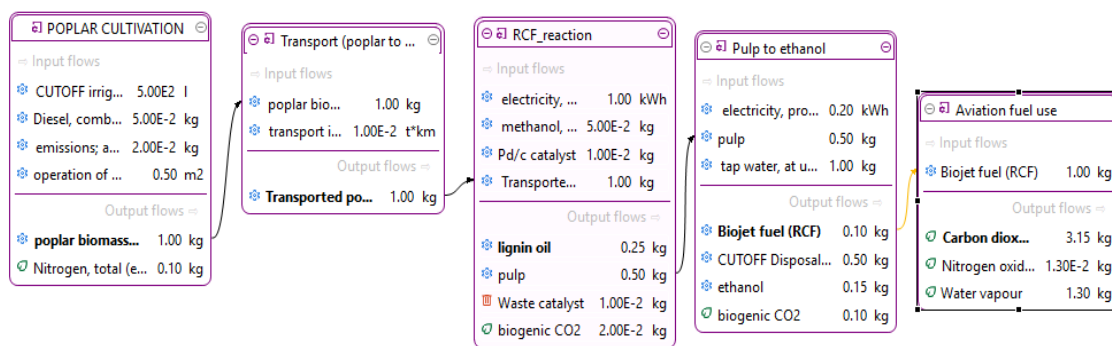


Figure 10: LCA Process flow diagram

9.2. Life Cycle Inventory (LCI)

Inputs per Functional Unit (1 kg poplar)

Input	Amount	Source / Justification
Poplar biomass	1.0 kg	Marginal land SRWC[54].
Methanol (solvent)	0.05 kg	90% recovery assumed[51].
Pd/C Catalyst	0.01 kg	Average reuse 5 times [52]
Electricity (US mix)	4.0 kWh	Includes RCF reactor, distillation, fermentation
Water	1.0 kg	Approximate for process and washing stages

Outputs

Output	Amount	Notes

Lignin oil (monomers + dimers)	0.25 kg	From RCF, ~50% monomers [51]
Carbohydrate pulp	0.50 kg	Hydrolyzed to sugars for fermentation
Ethanol	0.15 kg	From pulp via enzymatic hydrolysis [54]
Biojet fuel (ATJ)	0.10 kg	65% ethanol conversion efficiency [54]
Biogenic CO ₂ (fermentation)	0.10 kg	$C_6H_{12}O_6 \rightarrow 2 C_2H_5OH + 2 CO_2$
Biogenic CO ₂ (combustion)	0.315 kg	From 1 kg jet fuel combustion [54]
Spent catalyst	0.002 kg	Sent for metal recovery [54]

9.3. Life Cycle Impact Assessment (LCIA)

Method Used: ReCiPe Midpoint (H) 2016

Impact Category	Value	What It Means
Climate Change (GWP100)	0.28 kg CO ₂ e	Total carbon footprint
Fossil Resource Scarcity	0.08 kg oil	How much fossil fuel equivalent is used

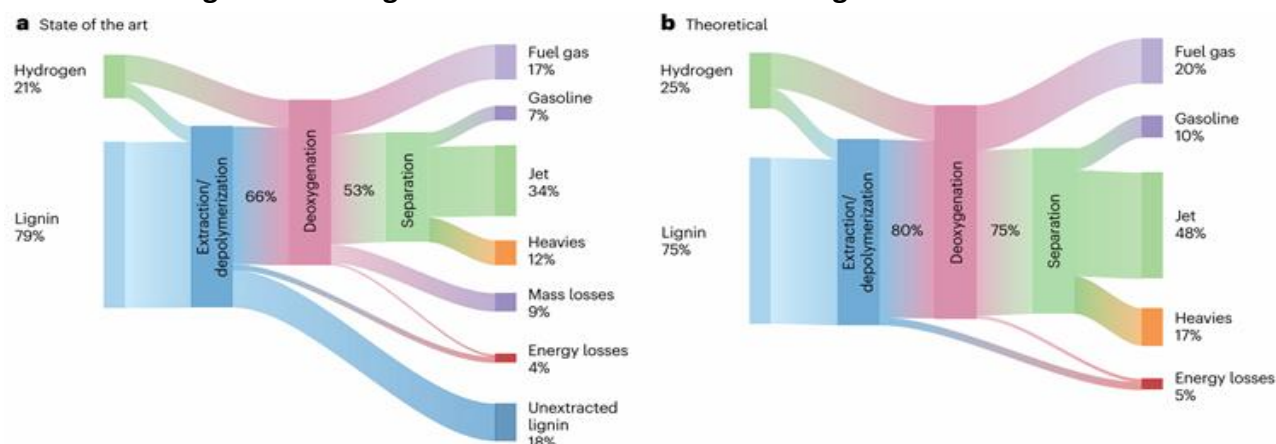
Human Toxicity	0.05 kg 1,4-DCB eq	Possible health impacts from chemical exposure
Terrestrial Ecotoxicity	0.004 kg 1,4-DCB eq	Toxic effect on land ecosystems
Water Use	1.2 m ³	Water consumed throughout the process
Particulate Matter Formation	0.002 kg PM10 eq	Air quality impact from fine particles

9.4. GWP Contribution by Process Stage (per 1 kg poplar biomass)

Process stage	Contribution (kg CO ₂ eq)	Percentage
Catalyst Production (Pd/C)	0.10	35.7%
Electricity Use (RCF + ATJ)	0.08	28.6%
Methanol Production	0.06	21.4%
Transport & Cultivation	0.03	10.7%
Others (misc. emissions, water)	0.01	3.6%

Total GWP100 = 0.28 kg CO₂ eq

10. Advantages of lignin first substrate to generate SAF blendstocks



The use of the lignin first method, particularly the Reductive catalytic fractionation (RCF) has emerged as a promising way to generate sustainable aviation fuel (SAF) blend-stocks, overcoming limitations with traditional lignin feedstocks. Conventional processes usually rely on technical lignins or pyrolysis oils, which contain heavily degraded lignin complexes with large C-C interlinkage concentrations. These interlinkages limit depolymerization efficiency and reduce the yield of jet-range hydrocarbons, as evidenced in studies like those by Yang et al., where alkaline-extracted corn stover lignin showed poor jet fuel yields despite full deoxygenation using Ru/Al₂O₃ and H-FAU catalysts. Nuclear magnetic resonance analysis confirmed the structural limitations of these lignins, and the resulting product oils contained high levels of complex bicyclo- and tricyclo-alkanes, which restrict their compatibility with aviation fuels.

In contrast, lignin-first strategies that stabilize lignin during extraction preserve more reactive ether linkages, allowing higher jet-range yields without needing extensive C–C bond cleavage catalysis. Recent advancements in this area, particularly by Stone et al., demonstrated the potential of using neat poplar RCF oil in a continuous trickle-bed reactor with a Mo₂C catalyst. This process achieved a 34% enthalpic yield of jet-range products based on the lignin content in planta. As illustrated in Figure 6a, this experimental setup shows the flow of combustion enthalpy through the system. However, this yield remains below theoretical potential due to incomplete delignification—only ~70% oil yield was achieved using methanol as a solvent, which lacks sufficient polarity and acidity for full lignin extraction.

Figure 6 above presents a theoretical energy flow scenario assuming complete delignification and total mass closure, which could significantly enhance the efficiency of the process. Improvements such as switching to alcohol/water solvent mixtures, which offer better delignification due to higher polarity and Lewis acidity, could increase the oil yield and thus jet-fuel production. Additionally, minimizing energy and mass losses through better product capture and reduced coke formation would allow for greater energy retention. Theoretical models suggest that, with optimization, the process could retain up to 48% of the original lignin combustion energy, comparable to established SAF production methods like ethanol-to-jet (ETJ) and hydro-processed esters and fatty acids (HEFA), which retain about 50–52%. Further enhancement through feedstock engineering or introducing C–C bond-cleaving chemistries like hydrocracking may also reduce heavy fraction formation and improve overall product quality

11. Conclusions and Outlook

A conceptual framework for converting lignin to sustainable aviation fuel (SAF) is illustrated in Figure 7, showing a sequence of reductive catalytic fractionation (RCF), deoxygenation, and distillation. This system highlights four critical research directions: feedstock modification, lignin fractionation enhancement, catalyst development, and process by-product utilization. Ensuring long-term catalyst and process stability is also emphasized as a key factor for commercial viability, given the harsh and complex nature of real biomass feeds.

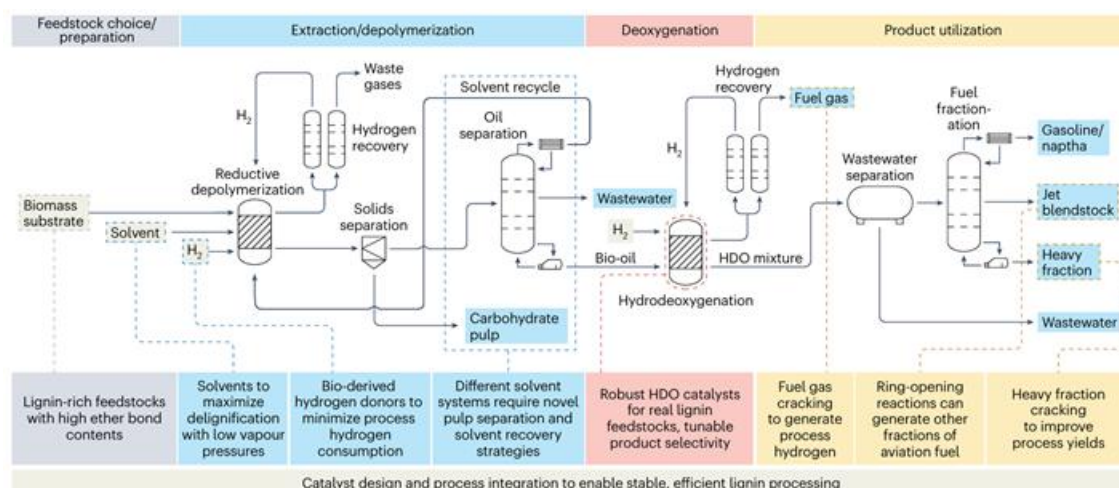


Fig. 7 | Opportunities for developing a viable lignin-to-SAF biorefinery. A simplified process flow diagram highlighting important research directions and process opportunities to improve a potential biorefinery for the generation of SAF from lignin.

Figure 11: Process flow diagram of a JET Fuel producing Biorefinery

Feedstock selection and preparation are pivotal to maximizing SAF yields. Lignins with native-like structures are easier to depolymerize and process than heavily crosslinked (C–C coupled) technical lignins. C-lignin is a high-energy, dense, linear feedstock, a promising feedstock for SAF production. Strategies that merge C-lignin into high-yielding biomass or application of milder extraction like enzymatic hydrolysis with minimal pre-treatment, can help preserve lignin’s native structure and enhance process efficiency.

Extraction and depolymerisation represent important barrier, as incomplete delignification hampers SAF yields. For lignin-first strategies to be effective, it is essential to engineer extraction systems that maximize delignification while operating under milder, lower-pressure conditions. Although traditional RCF relies on volatile solvents and high pressures, alternative non-volatile or bio-derived solvents may increase delignification and enhance bio-oil quality. However, this shift would require new separation technologies since products may no longer be recoverable via simple distillation. Incorporating bio-based hydrogen donors into these systems can further reduce fossil hydrogen demand and enhance the renewable content of the final SAF product. In the deoxygenation stage, catalyst design becomes central. For industrial deployment, hydrodeoxygenation (HDO) catalysts must achieve deep oxygen removal (<0.5 wt%) while allowing tunable hydrogenation pathways—producing a desirable mix of aromatic and cyclic alkanes. Moreover, these catalysts must resist deactivation caused by impurities or heavy

molecules in real biomass streams. Future research should prioritize the identification and scale-up of such robust, selective catalysts.

Process by-product utilization offers significant opportunities to improve overall energy efficiency. Methane, often generated through de-methoxylation during deoxygenation, could serve dual roles—as internal process fuel or as a hydrogen source via reforming. This internal hydrogen loop could increase the percentage of combustion energy retained from lignin. Additionally, heavier fractions of the product stream could be cracked into more jet-range molecules, while cyclic products could be ring-opened and isomerized into iso and n-alkanes. These pathways move toward the possibility of producing 100% lignin-derived jet fuel. Co-processing lignin with feedstocks like fatty acids may also simplify deoxygenation steps and broaden the product range.

The lifetime and durability of catalysts and processes will define the commercial success of lignin-to-SAF systems. Although progress has been made using model substrates, challenges remain in understanding how real feedstocks affect catalyst structure and activity over time. Addressing this needs innovations in catalyst regeneration (coke removal, acid washing, site repair) and improved process design to mitigate contaminants like salt and heavy carbonaceous deposits. Industry and academia must work closely together to convert lab-scale findings into a scalable biorefinery system.

In conclusion, lignin provides a sustainable route to aviation's objective of net-zero emissions by 2050, notwithstanding the fact that this is an ambitious target. As feedstock engineering, solvent systems, catalysis, and process integration continue to advance, lignin valorisation may be able to revolutionise the manufacture of large quantities of renewable SAF from a plentiful but underutilised biomass source.

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